COMPLETE PROJECT DOCUMENTATION

UIU ResearchCollab

Connecting Minds. Creating Research.
Student + Faculty + Admin collaboration portal

Discover • Team up • Get mentored • Publish

PROJECT TYPE	Role-based web portal (Core PHP + MySQL)
PORTALS	Student | Faculty | Admin
STACK	PHP 8 (PDO), MariaDB/MySQL, Bootstrap 5.3.3, vanilla JS
SCALE	127 PHP files, 22,000+ lines, 57 database tables
ORIGINAL REPO	github.com/AbrarAdib18/UIU-Research-Collaborator-
INSTITUTION	United International University (UIU), Dhaka
DOCUMENT	Version 1.1 - 9 November 2026

Prepared from the complete project source code, database scripts and files.

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1. Executive Summary

UIU ResearchCollab is a complete academic research-collaboration platform for United International University. It lets **students** build research profiles, find like-minded collaborators, apply to faculty-posted research opportunities, form teams with a private workspace and request a **faculty advisor**; lets **faculty** publish opportunities, review applicants, mentor students and teams; and gives **administrators** full oversight: user management, faculty verification, master data, moderation, reports and platform settings.

It is written in **Core PHP with PDO and MySQL/MariaDB** - deliberately no framework - with Bootstrap 5 and vanilla JavaScript on the front end, and runs on XAMPP. According to the project's own test report, 61 pages were swept over real HTTP against a real database with zero PHP errors.

3	127	22k+	57	15
Portals	PHP files	Lines of PHP	DB tables	Seed users

Highlights

- **Smart matching** - weighted 0-100 compatibility score (domains, skills, department, level, availability).
- **Full mentorship workflow** - typed advisor requests, transactional acceptance, capacity checks, team advisors.
- **Team workspace** - tasks, milestones, secure files, messages, invitations and join requests.
- **Communities, repository and saved items** - discussion groups and a shared research-resource library.
- **Direct chat** between students and faculty (4-second AJAX polling with JSON API).
- **Admin governance** - 28 admin pages, activity audit log, CSV reports and 7 live platform settings.
- **Security by design** - prepared statements, CSRF tokens, hashed passwords/reset tokens, role guards, upload hardening.

2. Problem Statement and Objectives

Research at a university often happens by chance: students do not know who shares their interests, faculty opportunities are announced informally, final-year project supervision is arranged through personal contacts, and nothing records collaboration history. The result is missed partnerships and uneven access to mentors.

#	Objective	Delivered by
1	Make researchers discoverable	Research Connect directory with filters and match scores; researcher profiles with visibility control.
2	Open research opportunities fairly	Faculty-created opportunities, student applications, shortlist / accept / reject with notifications.
3	Enable teamwork	Teams with tasks, milestones, files, messages, invitations, join requests and size limits.
4	Structure mentorship	Advisor requests (6 types), assignments, feedback, faculty capacity rules.
5	Build community and knowledge sharing	Public / private communities, posts, comments, research repository, saved items.
6	Govern the platform	Admin portal: verification, moderation, logs, reports, settings.
7	Protect users and data	Role-based access, privacy toggles, audit trail, safe uploads.

3. Users and Portals

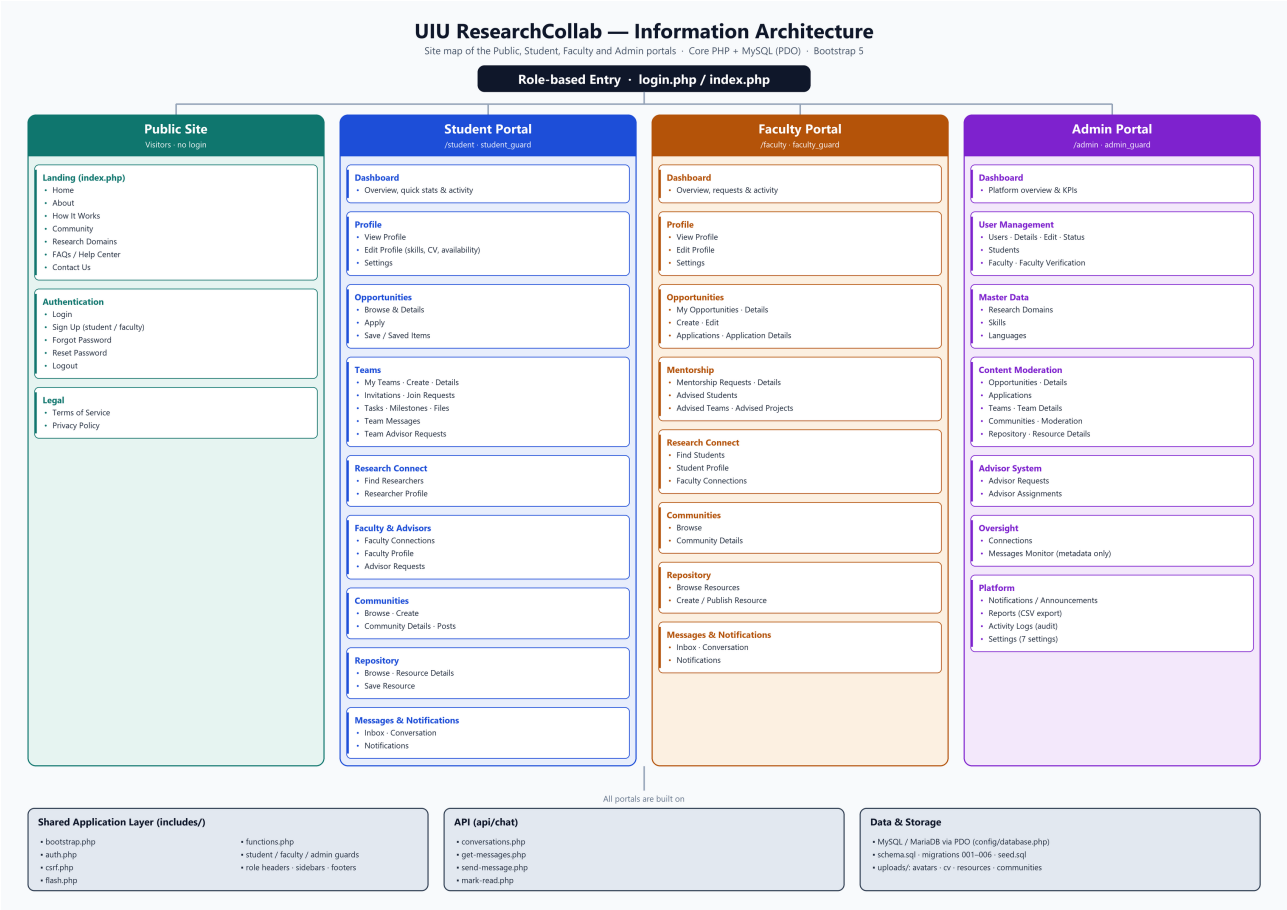


Figure 1 - Information architecture: public site and the three role portals (from the project's docs folder).

Portal	Audience / entry	Pages	Core capabilities
Public site	Visitors; index.php	index, about, how it works, domains, FAQ, contact, terms, privacy, login, signups, password reset	Learn about the platform, register, recover password, send contact message
Student	role = student; student/	34	Profile, Research Connect, opportunities, teams, communities, repository, saved items, advisor requests, messages, notifications, settings
Faculty	role = faculty; faculty/	25 (17 functional modules)	Profile, opportunity CRUD, applications review, mentorship inbox, advised students/teams, connections, chat, repository, communities
Admin	role = admin; admin/	29 (28 modules)	Users, faculty verification, master data, moderation, advisor oversight, announcements, logs, reports, settings

4. Features in Detail

4.1 Authentication and account safety

- **Student sign-up** validates duplicate e-mail and student ID and, by default, the university domain (@bscse.uiu.ac.bd - the domain is an admin setting). **Faculty sign-up** (faculty-signup.php) requires any @uiu.ac.bd address, department, designation, optional faculty ID, and a password of 8+ characters with a special character.
- Passwords use `password_hash()`; sessions are regenerated on login; inactive or suspended accounts are blocked.
- **Mid-session protection:** `current_user()` re-reads the account status on every request, so an admin suspension logs the user out at once.
- **Forgot / reset password:** single-use token, 1-hour expiry, stored as a SHA-256 hash. With no mail server configured, the reset link is shown on screen (demo convenience).
- **Role-based routing:** login sends each role to its own dashboard; `require_student/faculty/admin` guards protect every page.

4.2 Student portal

Module	What it does
Dashboard	Profile completion %, counts of domains / skills / applications / teams, unread notifications, recommended collaborators, latest opportunities, upcoming milestones, activity feed, advisor and message counters.
Profile	Personal, contact and academic data; domains and skills with levels; education; preferences and methodologies; availability schedule; extracurriculars; projects, publications, certifications, achievements, languages, work experience; photo, cover and CV upload; 4 privacy toggles; URL scheme validation.
Research Connect	Searchable, filterable, paginated directory (domain, skill, department, level) with match-score badges and a "How it works?" modal; read-only researcher profile; Invite to Team. A second view finds faculty (filter: accepting mentees).
Opportunities	Browse, filter, details, apply with message, withdraw, re-apply, save; duplicate, deadline and status enforcement; notifications.
My Teams	Create (size limit), discover open teams, request to join, invitations, workspace tabs: tasks, milestones, files (membership-gated download), messages; faculty-advisor panel.
Communities	Create (creator becomes Admin member), browse, join / leave, posts and comments, private communities enforced server-side.
Repository / Saved	Browse by type, year, domain; add file or link; edit own; save resources; unified saved view with opportunities.
Advisors and messages	Faculty profile with Request Advisor; request tracking, cancel, answer clarification; connections; direct chat with unread badge.
Notifications / Settings	Deep-linked notifications with mark-read; change password; privacy toggles.

4.3 Faculty portal

Module	What it does
Dashboard	Active opportunities, pending applications and mentor requests, advised students / teams, connections, unread items, upcoming advised-team milestones.
Opportunities	Create / edit / view with status flow Draft → Open → Closed → Completed, domain tagging, ownership checks; unverified faculty cannot publish when verification is required.
Applications	Cross-opportunity inbox, applicant profile (visibility-respecting), internal review notes, Shortlist / Accept / Reject with student notifications.
Mentorship	Inbox with filters; Accept (transactional), Decline with reason, Request Clarification; advised students, teams (workspace with guidance posts) and projects; mark Completed.
Research Connect	Find students and teams, filter by domain / department / skill / seeking-advisor; Connect or Message.
Other	Communities, repository, connections, messages, notifications, settings with verification status.

4.4 Admin portal

Area	Capabilities
Users	Search / filter / sort; activate, deactivate, suspend, restore; edit safe fields only; cannot act on self or deactivate the last active admin.
Faculty verification	Pending queue; Approve, Reject with reason, Request update; transactional; notifies the faculty member.
Master data	CRUD for research domains, skills, languages; duplicate prevention; delete blocked while referenced.
Moderation	Opportunities (close, reopen, archive, delete), applications (mark invalid), teams (archive, restore, remove member), communities (suspend, restore, delete, hide posts and comments), repository (publish, hide, delete).
Advisor oversight	Platform-wide requests and assignments; mark invalid; end assignment; capacity warning panel.
Oversight	Connections (block); chat monitor shows **metadata only** (participants, dates, counts) - never message content.
Communication	Own inbox and announcement composer (all / students / faculty / selected e-mails).
Audit and reports	Searchable activity log; reports as progress-bar breakdowns with date-range counts and CSV export (4 datasets, CSV-injection guarded).
Settings	7 settings with real effect (Table below).

Setting key	Effect
`public_registration_enabled`	Turns self-registration on / off for students and faculty
`student_email_domain`	Domain pattern accepted at student sign-up
`default_profile_visibility`	Visibility given to new profiles
`default_team_size_limit`	Pre-filled team size in Create Team
`faculty_verification_required`	Unverified faculty cannot publish opportunities
`community_creation_policy`	open or admin_approval (new community created Inactive until approved)
`maintenance_notice`	Banner shown on public pages when non-empty

5. Key Algorithms and Business Rules

5.1 Research Connect matching score

For viewer V and candidate C, `calculate_match_score()` returns a score from 0 to 100. It is reused by Research Connect and by the Dashboard's Recommended Collaborators, and explained to users in a modal.

Factor	Weight	Rule
Shared research domains	40	(shared domains / V's domain count) x 40
Shared skills	30	(shared skills / V's skill count) x 30
Department / program	15	+10 same department, +5 more if same program
Academic-level closeness	10	From trimester number: +10 within 1, +5 within 3, else 0
Mutual availability	5	+5 if both are looking for a team

Example: V has 4 domains and 3 skills. C shares 3 domains and 2 skills, same department but different program, 2 trimesters apart, both looking for a team: 30 + 20 + 10 + 5 + 5 = **70%**.

5.2 Advisor request workflow

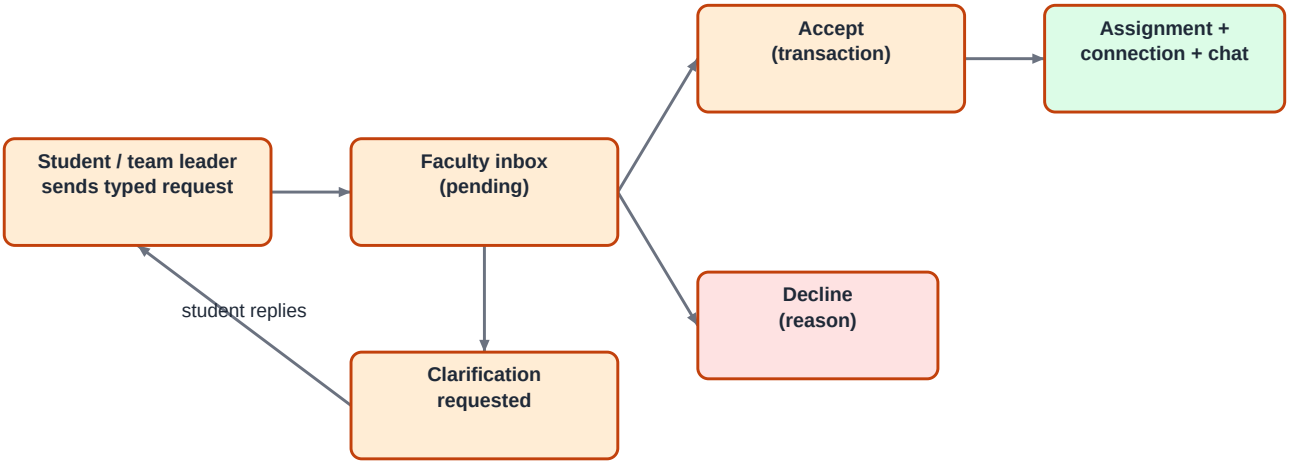


Figure 2 - Advisor request lifecycle. A clarification reply returns the request to pending.

- Request types: research_advisor, paper_advisor, project_mentor, fydp_supervisor, research_mentor, technical_mentor.
- Accept checks faculty capacity and duplicate active assignments **inside a database transaction**; it then updates the request, inserts the assignment, creates or reuses an accepted connection and conversation, and notifies the student or every team member.
- Duplicate pending requests are rejected (the schema notes MariaDB 10.4 lacks partial unique indexes, so the rule is enforced in code).
- Faculty may post team guidance (advisor_feedback) that notifies all active members.

5.3 Connections and chat

- Either role sends a connection request; unordered-pair duplicates are blocked by a generated `pair_key` column and self-connection by a CHECK constraint.
- `can_message()` allows chat only if the users have an accepted connection *or* an active advisor assignment.

- Chat API in `api/chat/`: conversations, get-messages (since-last-id), send-message (POST, CSRF, 2000-char cap), mark-read - JSON with 401/403/405/500 codes.
- The page works without JavaScript (server-rendered history) and adds fetch() polling every 4 seconds, paused when the tab is hidden.
- Notifications are aggregated per thread, avoiding one notification per message.

6. System Architecture

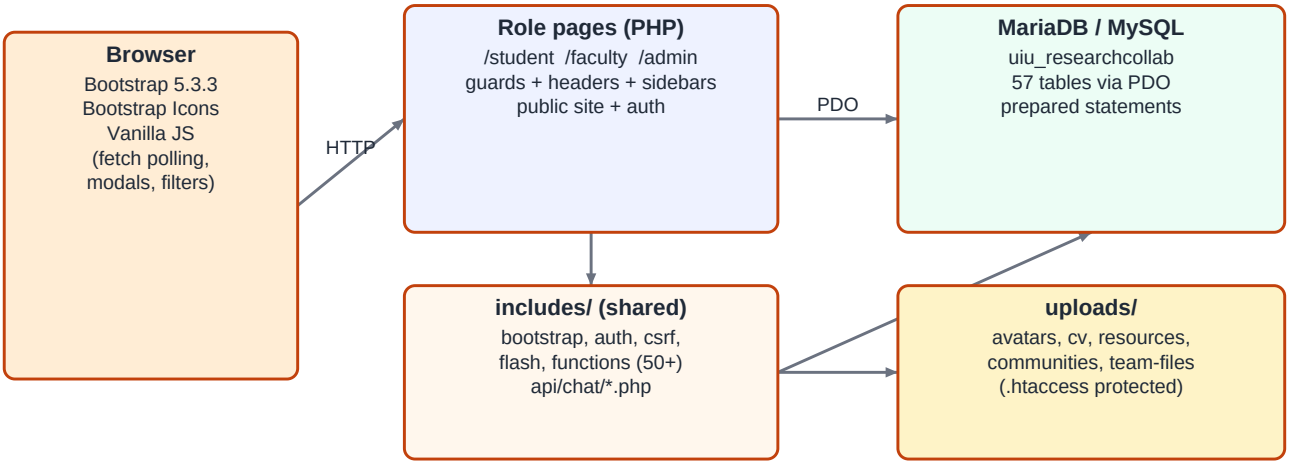


Figure 3 - Layers and data flow.

Layer	Technology	Notes
Front end	HTML5, CSS3 (CSS/ design system), Bootstrap 5.3.3, Bootstrap Icons, vanilla JavaScript	Original static HTML mock-ups (Index, Login, Signup, Dashboard, Profile, Research-Connect .html) preserved as the design source
Back end	Core PHP 8 (no framework), PDO with exceptions, emulated prepares off	includes/functions.php holds 50+ helpers; bootstrap forces display_errors off
Database	MariaDB 10.4 / MySQL, utf8mb4, InnoDB, FKs, CHECKs, generated columns	schema.sql + 6 migrations + seed.sql
Server	XAMPP (Apache, PHP 8.2, phpMyAdmin)	Auto-detects its base URL; .htaccess guards uploads
PHP extensions	pdo_mysql, mysqli, mbstring, fileinfo, curl, gd	gd is disabled by default in XAMPP and must be enabled

6.1 Folder structure

Path	Content
`/ (root)`	index.php, login.php, signup.php, faculty-signup.php, logout.php, forgot/reset-password.php, contact.php, faq.php, terms.php, privacy.php; static design HTML
`config/`	database.php - PDO connection (XAMPP defaults)
`includes/`	17 shared files: bootstrap, auth, csrf, flash, functions; header / sidebar / guard per role; public header and footer
`student/ , faculty/ , admin/`	34, 25 and 29 pages
`api/chat/`	4 JSON endpoints shared by both chat UIs
`database/`	schema.sql, migrations 001-006, seed.sql
`docs/`	information-architecture.png, user-journey.png and the Python scripts that generate them
`CSS/ , JS/ , IMAGES/`	Design system (dashboard, profile, login, signup, research-connect, style), app.js, images
`uploads/`	avatars, cv, resources, communities, team-files (content git-ignored)

7. Database Design

The database **uiu_researchcollab** has **57 tables**: 35 in the base schema and 22 added by migrations (repository, password reset and contact, extended profile, faculty portal, admin portal). Migrations are idempotent and must be run in order.

Domain	Tables
Identity	users (role student|faculty|admin; status active|inactive|suspended), password_reset_tokens, platform_settings, activity_logs, notifications, contact_messages
Student profile	student_profiles, profile_research_domains, profile_skills, profile_languages, profile_visibility, research_preferences, education, projects, publications, certifications, achievements, work_experience, extracurricular_activities, profile_availability
Master data	research_domains, skills, languages
Opportunities	research_opportunities, opportunity_domains, opportunity_applications (status incl. Shortlisted), saved_opportunities
Teams	research_teams, team_members, team_invitations, team_requests, team_tasks, team_milestones, team_files, team_messages
Communities	communities, community_members, community_posts, community_comments (is_hidden for moderation)
Repository	research_resources, saved_resources
Faculty	faculty_profiles, faculty_research_domains, faculty_skills, faculty_education, faculty_publications, faculty_projects, faculty_availability, faculty_preferences, faculty_visibility, faculty_verifications
Mentorship	advisor_requests, advisor_assignments, advisor_feedback
Messaging	research_connections (pair_key), direct_conversations, direct_messages

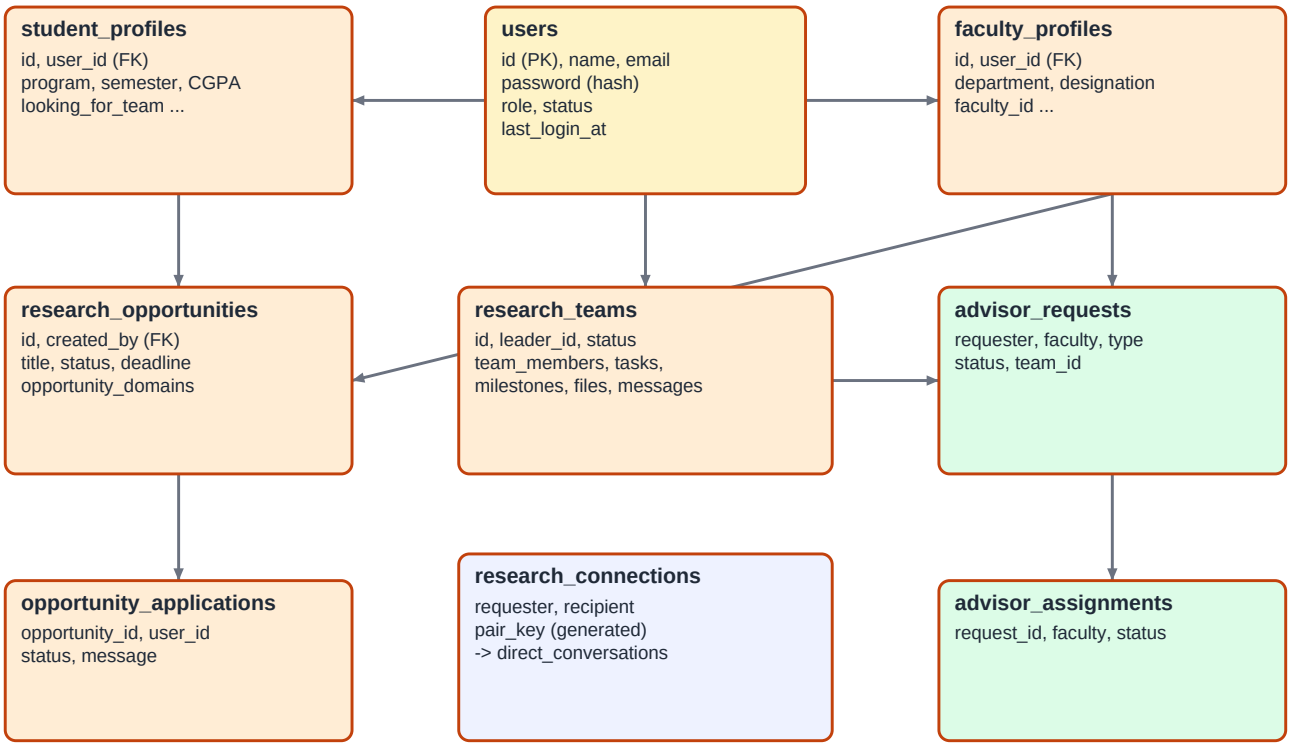


Figure 4 - Simplified relationships (core tables only).

7.1 Notable design decisions

- **Reuse of status ENUMs** for moderation (users.status, research_opportunities.status, research_teams.status, etc.) - no extra moderation columns; reasons go to activity_logs.description.
- **Generated column** pair_key = LEAST(a,b)-GREATEST(a,b) gives O(1) duplicate detection for unordered pairs.
- **CHECK constraints** block self-connections and self-conversations.
- **Announcements reuse notifications**; no extra table.
- **Normalised faculty profile** mirrors the student profile structure so shared helpers and CSS work for both.

7.2 Representative rule: capacity check

```
-- advisor accept (simplified)
BEGIN;
  -- 1. faculty_capacity_remaining() > 0 ?   2. no active assignment for requester ?
  UPDATE advisor_requests    SET status='accepted' WHERE id=? AND status='pending';
  INSERT INTO advisor_assignments (...) VALUES (...);
  INSERT INTO research_connections (...) ON DUPLICATE ... (accepted);
  -- get-or-create direct_conversations (linked to connection + assignment)
COMMIT;
```

8. Security Review

Control	Implementation	Evidence
SQL injection	PDO prepared statements everywhere	config/database.php (ERRMODE_EXCEPTION, no emulation)
CSRF	csrf_field() / require_csrf() on every state-changing POST and chat API	includes/csrf.php
Passwords	password_hash / verify; reset tokens SHA-256 hashed, single use, 1 h expiry	includes/auth.php
Sessions	session_regenerate_id on login; account status re-checked every request	current_user()
Authorisation	Role guards; ownership checks in SQL (WHERE id=? AND user_id=?); team membership and advisor checks; can_message()	guards + helpers
Privacy	profile visibility plus 4 granular toggles for both students and faculty; private communities gated server-side; admin chat monitor shows metadata only	profile_visibility, faculty_visibility
Uploads	Extension allow-list, dangerous-extension block, size limit, random names, is_uploaded_file; team files served only via a membership-gated route; .htaccess returns 403 on direct access	validate_upload(), safe_filename()
Output / links	e() htmlspecialchars everywhere; javascript: URLs rejected	is_valid_http_url()
Errors	display_errors forced off; generic 'Service temporarily unavailable'	includes/bootstrap.php
CSV export	Cells guarded against formula injection	admin/reports.php

Residual risks and recommendations

Priority	Item	Recommendation
Medium	Team-capacity check at accept time is not row-locked; two simultaneous accepts could both pass.	Wrap in SELECT ... FOR UPDATE inside the transaction.
Medium	No outbound e-mail: reset link is displayed on screen.	Integrate SMTP and remove on-screen links before production.
Medium	Demo credentials and default root DB user (blank password) are for local use only.	Use per-environment config (database.local.php) and strong secrets; never deploy seed accounts.
Low	No automated test suite; verification was manual and scripted curl / PHP.	Add PHPUnit and a CI job running the 61-page sweep.
Low	Responsive and pixel-level browser QA was never automated.	Add Playwright or Cypress smoke tests.
Low	Match scoring runs in PHP over the current page.	Move to SQL or a cached job at larger scale.

9. Installation and Run Guide

1. Install **XAMPP** (PHP 8+). Enable the **gd** extension in php.ini (`extension=gd`) and restart Apache.
2. Copy the project into `C:\xampp\htdocs\UIU-ResearchCollab-main` (any folder name works; base_url() auto-detects).
3. Start Apache and MySQL. In phpMyAdmin create database `uiu_researchcollab` (utf8mb4, utf8mb4_unicode_ci).
4. Import in this exact order: `schema.sql`, `migrations.sql`, `migrations_002_landing.sql`, `migrations_003_profile_extended.sql`, `migrations_004_communities.sql`, `migrations_005_faculty_portal.sql`, `migrations_006_admin_portal.sql`, `seed.sql`. Run seed.sql only once on a fresh database. (An older full phpMyAdmin dump that predates the migrations is not included, because it would create an incomplete database.)
5. Defaults in config/database.php need no change on stock XAMPP (127.0.0.1, root, empty password). Otherwise create config/database.local.php.
6. Make sure uploads/ sub-folders are writable by Apache.
7. Open `http://localhost/UIU-ResearchCollab-main/`.

Demo accounts (local only)

Role	E-mail	Notes
Student	student@example.com	8 students seeded (e.g. tanvir.ahmed@bscse.uiu.ac.bd)
Faculty	faculty@example.com	5 faculty; one not accepting mentees; nasrin.akter@... is pending verification
Admin	admin@example.com	admin2@example.com enables the last-active-admin rule demo

The seed password is documented in docs/DEVELOPMENT_LOG.md (the original project README); it is intended for local demonstration only.

10. User Journey and Testing

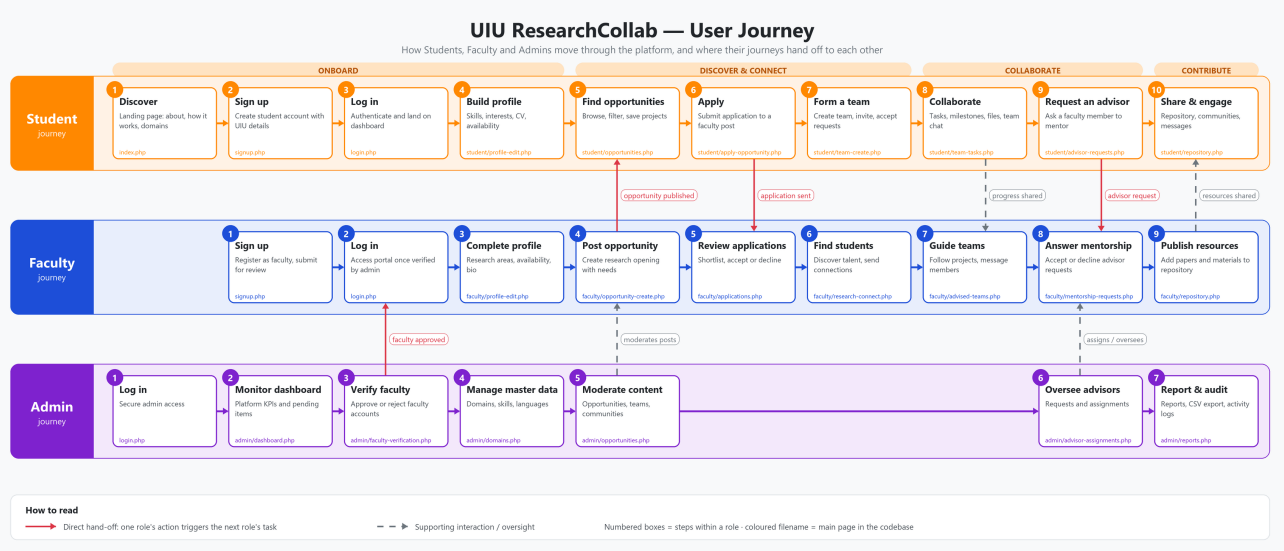


Figure 5 - End-to-end user journey (from the project's docs folder).

10.1 Test cases

ID	Area	Steps	Expected
T1	Sign-up rules	Student sign-up with a non-UIU domain	Rejected with a clear message
T2	Role routing	Log in as each role	Lands on /student, /faculty, /admin dashboard
T3	Suspension	Admin suspends a logged-in student, student clicks any link	Forced logout with notice
T4	CSRF	POST a form without a token	Request refused
T5	Match score	Open Research Connect, click How it works	Badges and weighted explanation shown
T6	Apply	Apply, withdraw, re-apply to an opportunity	Status changes; duplicate blocked; notification sent
T7	Team privacy	Non-member opens team-files / tasks URL	Redirected; private tabs hidden
T8	File download	Open an uploads/team-files URL directly	403 Forbidden; member route returns the file
T9	Private community	Non-member opens a private community URL	Private notice, no post content
T10	Advisor accept	Faculty accepts a pending request	Assignment, connection and conversation created; student notified
T11	Capacity	Accept when mentee capacity is full	Blocked with an error
T12	Chat	Student and faculty exchange messages	Appear within ~4 s; unread badge updates
T13	Admin protection	Deactivate the only active admin	Refused
T14	Settings	Turn registration off	Sign-up pages refuse new accounts
T15	CSV	Export a report	File downloads; cells beginning with = are neutralised

Reported verification

The project development log (docs/DEVELOPMENT_LOG.md) reports a runtime sweep of 61 pages with zero PHP errors against a fresh XAMPP MariaDB import (all migrations and seed applied without errors), plus live checks of upload rejection, URL-scheme validation, 403 on protected folders and cross-user notification tampering. No automated suite exists.

11. Known Limitations and Roadmap

- Opportunities can be created only by faculty / admin (no student-created projects).
- Team messaging is page-refresh based; only faculty-student chat uses polling.
- No calendar or events module, no outbound e-mail, no payments, no newsletter, no content-report (flagging) system.
- Seeded team files reference placeholder names that 404 until a real upload.
- Section 10 of the development log still says faculty accounts are seed-only, but the delivered code contains faculty-signup.php (with UIU e-mail validation); update the log.
- Reference data (domains, skills) is managed by admins; no bulk import.

Phase	Improvement	Benefit
1	PHPUnit + browser tests, SMTP e-mail, row locking on accept	Reliability and real notifications
2	WebSocket or SSE chat for teams and direct messages	True real-time collaboration
3	Calendar (meetings, deadlines), reminders	Planning
4	Content reporting and moderation queue	Safer communities
5	Analytics dashboard, ML-assisted matching, publication tracking (DOI / ORCID)	Smarter discovery and impact
6	REST API, mobile app, SSO with UIU accounts	Reach and integration

12. What We Built - Team and Deliverables

Team member	Responsibility
Abrar Adib (GitHub: AbrarAdib18)	Owner of the original repository **github.com/AbrarAdib18/UIU-Research-Collaborator-** and author of all 6 commits in its Git history
Kazi Shibli Zaman Rahil (GitHub: Rahil03-git)	[add your role]
[Other team members]	[add names and roles]

Deliverable	Detail
Application	127 PHP files (about 22,300 lines): 34 student, 25 faculty, 29 admin pages, 17 includes, 4 API endpoints, public site
Database	57 tables, 7 SQL scripts, seed with 15 users and realistic demo data, full dump
Design system	6 CSS files, original HTML mock-ups, 6 images
Documentation	Development log (the original README) with dated reports (setup, XAMPP runtime, profile, communities, faculty and admin passes), IA and journey diagrams with generator scripts
Quality work	Fixed real bugs: role redirect, stale-session status check, community creation, upload folder protection

13. Presentation Guide and Viva Questions

13.1 Suggested 10-minute demo

1. Landing page and domains; sign-up rules (1 min).
2. Student dashboard and profile completion (1 min).
3. Research Connect, How-it-works modal and match percentages (1.5 min).
4. Apply to an opportunity; show status pill (1 min).
5. Team workspace: upload a file, show 403 on the direct URL (1.5 min).
6. Request an advisor; switch to faculty and accept; open chat (2 min).
7. Admin: verify faculty, hide a post, change a setting, export CSV (1.5 min).
8. Roadmap (0.5 min).

13.2 Likely questions and model answers

Q. Why Core PHP without a framework?

To show how routing, auth, CSRF, validation and PDO work from first principles; shared helpers in includes/ give framework-like structure.

Q. How is the match score calculated?

A weighted sum: 40 domains, 30 skills, 15 department / program, 10 level closeness, 5 availability, normalised to the viewer's own counts.

Q. How do you prevent two roles accessing each other's pages?

Every page starts with a guard (require_student / faculty / admin) that checks the session role; ownership and membership checks sit on top.

Q. Why a transaction when accepting an advisor request?

Several writes (request, assignment, connection, conversation) must succeed together or not at all, and capacity must be checked consistently.

Q. How does chat work without WebSockets?

Server-rendered history plus fetch() polling every 4 s against JSON endpoints, paused with the Page Visibility API.

Q. How is privacy protected?

Visibility toggles, private communities, membership-gated files, and an admin monitor that exposes metadata only.

Q. What does the generated pair_key column do?

It normalises (a,b) and (b,a) to the same key so an unordered connection pair is unique and fast to look up.

Q. How are uploads secured?

Allow-list, size limit, random filenames, is_uploaded_file, and .htaccess deny with an authorised download script for team files.

Q. What happens when an admin suspends a user mid-session?

current_user() checks status each request; a non-active account is logged out immediately.

Q. What are the limitations?

No e-mail, no real-time team chat, no automated tests, simple PHP-side scoring; all listed with fixes in Sections 8 and 11.

Q. How would you scale it?

Move scoring to SQL / cache, add a queue for notifications, introduce an API layer and a framework, and use WebSockets.

Appendix - Glossary

Term	Meaning
PDO	PHP Data Objects - database abstraction with prepared statements
CSRF	Cross-Site Request Forgery - prevented by secret per-session tokens
FYDP	Final Year Design Project
Polling	Client repeatedly asks the server for new data
Migration	Versioned SQL script that evolves the schema
Idempotent	Safe to run more than once with the same result

Term	Meaning
Verification (faculty)	Admin approval confirming a faculty identity